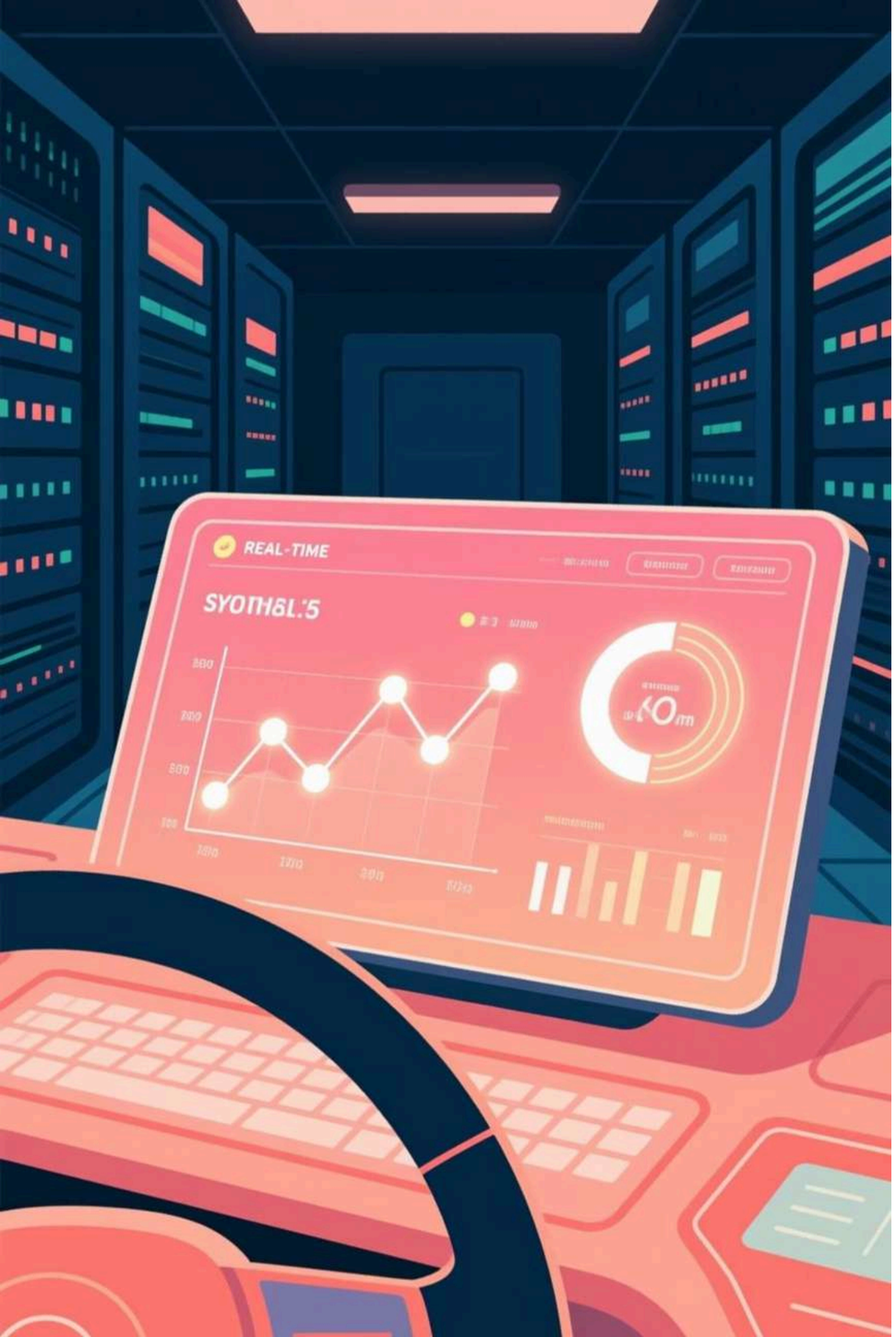


Real-Time System Resource Monitoring Dashboard

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Introduction: Empowering Developers and Admins

In the fast-paced world of system administration and development, having real-time insights into resource utilization is paramount. This dashboard, built entirely in Python, offers an intuitive and interactive way to monitor your system's heartbeat, ensuring optimal performance and proactive problem-solving.



Dashboard Architecture: A Seamless Flow



Python Backend

Core logic for data collection and processing.



Data Collection

psutil efficiently gathers system metrics.



Streamlit UI

Interactive and responsive web-based dashboard.



Plotly Visualization

Dynamic and insightful charts for data display.

Our dashboard leverages a robust, modular architecture, ensuring efficient data flow from system metric collection to dynamic visualization. This clear separation of concerns enhances maintainability and scalability, making it a powerful tool for any environment.

Key Dependencies: The Power Behind the Scenes

`streamlit`

The backbone for our interactive dashboard and user interface, enabling rapid development of data applications.

`psutil`

Cross-platform library for retrieving system utilization information, including CPU, RAM, disk, and process details.

`plotly`

Used for generating rich, interactive charts like time-series graphs, gauge meters, and bar charts for data visualization.

`pandas`

Essential for data manipulation and structuring, specifically for creating DataFrames and enabling CSV exports.

`requests`

Facilitates external API calls to fetch network information such as public IP address and ISP details.



Dashboard Layout: Intuitive & Comprehensive

Sidebar Controls

- Network Info: IP, City, Region, Country, ISP details.
- Settings: Adjustable refresh rate and historical data window.
- Unit Selection: Customize display units for various metrics.
- CPU Toggle: Switch between overall and per-core CPU usage.
- Start/Stop Buttons: Manual control over data collection.

Main Display Area

- CPU Usage: Real-time time-series chart.
- RAM Usage: Dynamic time-series chart.
- Disk Usage: Intuitive gauge chart for quick overview.
- Network Throughput: Live upload and download rates.
- Running Applications: Detailed table with process metrics.

The dashboard is designed for clarity and ease of use, segmenting controls and critical metrics into distinct, accessible areas. This layout ensures users can quickly find the information they need and customize their monitoring experience.



Feature Spotlight: Dynamic CPU Monitoring

Gain granular insights into your system's processing power.

Overall Usage

Visualize total CPU load over time with an interactive line chart.

Per-Core Breakdown

Toggle to view individual core performance, pinpointing bottlenecks.

Frequency & Count

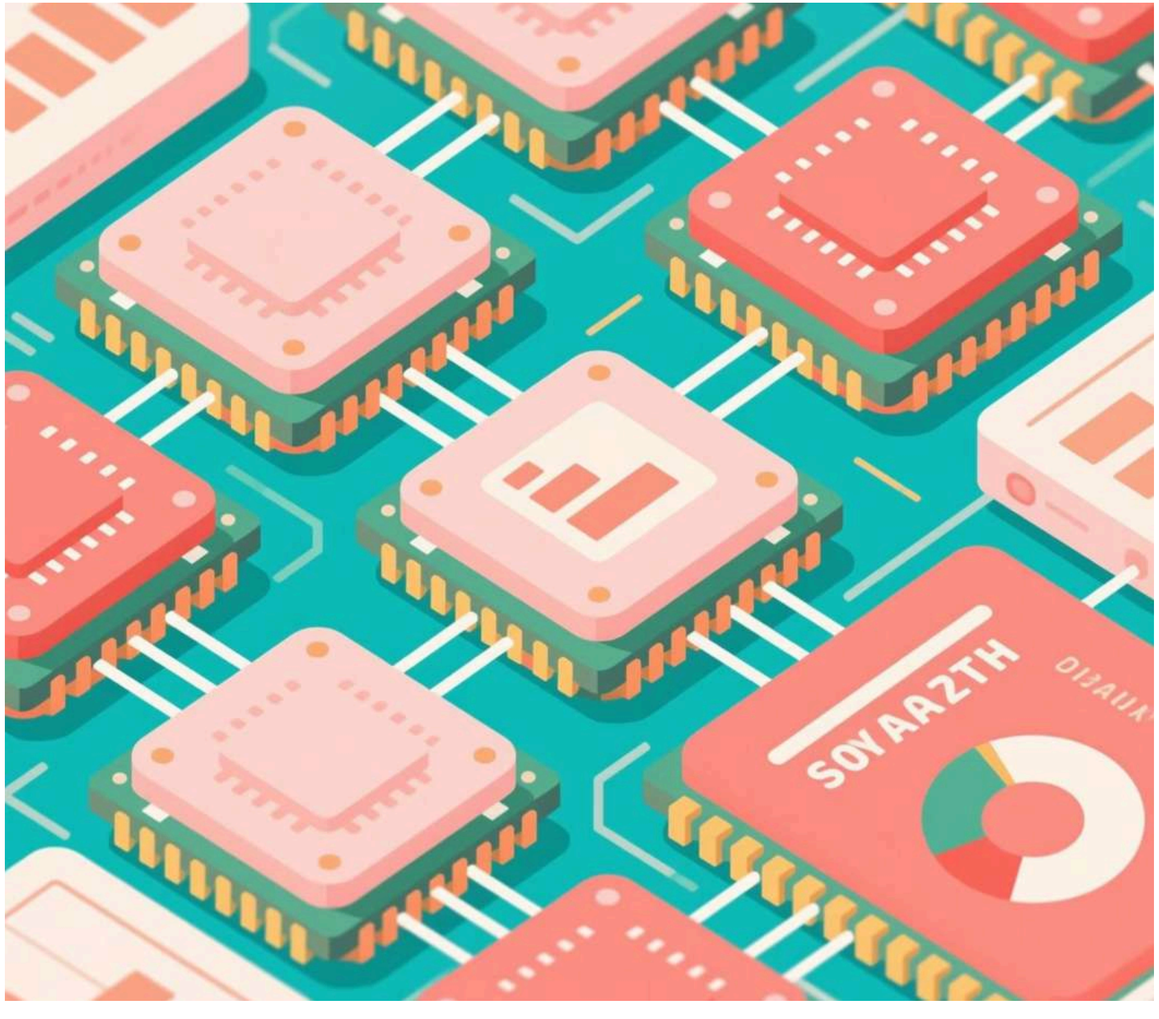
Display real-time CPU clock speed and the total number of logical cores.

Feature Spotlight: RAM & Disk Overview

Keep a close eye on your memory and storage health.

RAM Utilization

The dashboard provides a real-time percentage of RAM in use, alongside the total installed memory. A detailed chart visualizes memory consumption trends, helping identify potential memory leaks or applications hogging resources.



Disk Space Management

A prominent gauge chart immediately displays the percentage of disk space used, offering an at-a-glance status. You'll also see the total disk capacity, enabling proactive management to prevent storage-related performance issues.

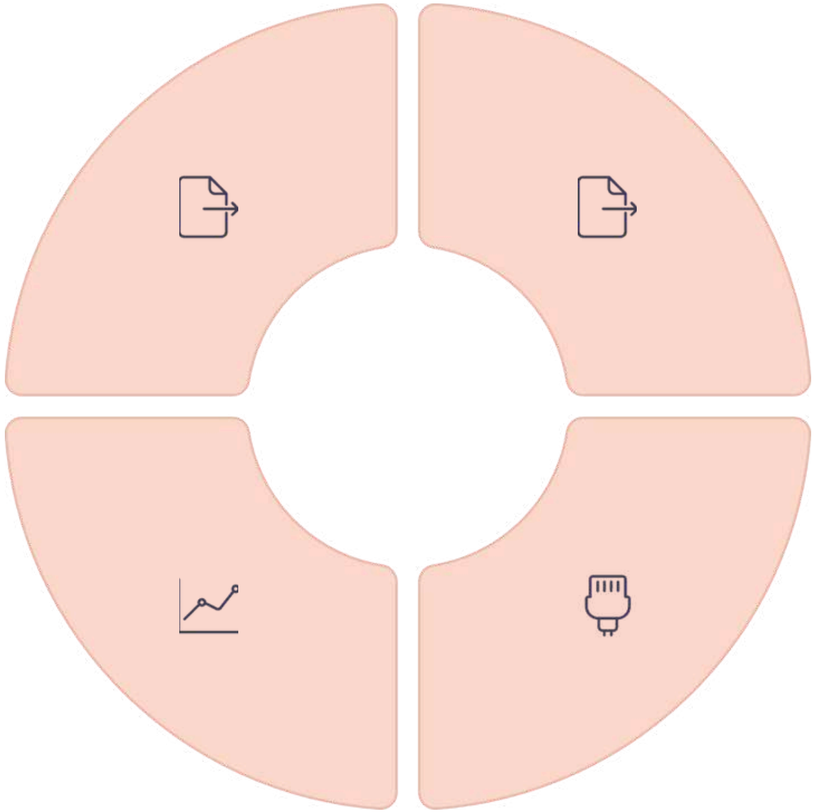


Feature Spotlight: Network Throughput Analysis

Monitor your internet activity with precision.

Upload Rate

Track real-time data being sent from your system.



Download Rate

Observe incoming data flow for internet speed assessment.

Trend Visualization

Time-series charts illustrate network usage peaks and valleys.

Flexible Units

View rates in KB/s, MB/s, or Mbps for tailored insights.

Understand your network's behavior with live KPIs and historical trends, crucial for identifying network saturation or unexpected data transfers.

Feature Spotlight: Comprehensive Application Tracking

Gain unprecedented visibility into your software usage patterns.

Running Applications List

See a detailed list of all currently active processes, helping you identify resource-intensive applications.

Times Opened

Track how many times each application has been launched, providing insight into usage frequency.

Screen Time Metrics

Monitor per-application screen time, revealing how much time is spent in specific programs.

Additionally, the dashboard aggregates total screen time across all applications, giving you a holistic view of system engagement. This feature is invaluable for productivity analysis and optimizing software workflows.



Feature Spotlight: Data Export & Reporting

Unlock the full potential of your monitoring data by effortlessly exporting historical system metrics. This feature provides a convenient way to analyze trends, generate reports, or integrate with other tools for deeper insights.

Timestamp
Exact time of each recorded metric.

CPU Usage
Percentage of processor utilization.

RAM Usage
Memory consumption in real-time.

Disk Usage
Storage allocation and free space.

Network Upload
Data sent from your system.

Network Download
Data received by your system.

All collected data points are formatted for easy analysis, allowing you to seamlessly integrate them into your preferred data management workflows.

Conclusion & Next Steps: Beyond Monitoring



Real-time & Interactive

Dynamic visualization for immediate insights.



Performance Analysis

Key for optimizing system resource usage.



Data Export Capability

History of metrics available in CSV format.

Future Enhancements:

- Alerting: Implement custom thresholds for notifications on critical resource levels.
- Logging: Integrate persistent data storage for long-term trend analysis.
- Cloud Integration: Expand monitoring capabilities to distributed or cloud environments.

Thank You!